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Chp No. 3

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Everyday Science

Chromosomes

Introduction

The nucleus of each cell contains a number of very small thread like

structure called chromosomes. The term *

chromosome was introduced into the scientific vocabulary by Waldeyer in 1889. Each chromosome consists of hundreds of molecules of nucleoproteins called genes.

In the nucleus of the cell, the chromosomes are present as two identical sets which pair off, each pair consisting of two identical chromosomes. Chromosomes are hereditary material.

Number and Types:

The number of chromosomes in each cell is fixed for a particular species. In human beings, it is 46. This is called as diploid number ($2n$). However, in spermatozoa and ova the number of

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Chromosomes is only half the diploid number, i.e. 23. This is called as Haploid number (n). Each of the 46 chromosomes is a member of a homologous pair, one member of each pair being received from the mother and one from the father.

Twenty-two of the pairs are identical in both sexes and are called Somatic chromosomes. In the male, there is one

'X' chromosomes and a clearly different chromosome, the 'Y' chromosomes. However, in the female, both the sex chromosomes are 'X' chromosomes. Chromosomes may

assume various shapes, twisted, spiral, Curved or rod-shaped.

Living Things

Chemical Composition:

The chemical constitution of chromosomes

are: (i) Deoxyribo Nucleic Acid (DNA)

(ii) Ribo Nucleic Acid (RNA)

(iii) Histones (iv) Acidic proteins

GENES

★ Introduction:

Genes are natural units of hereditary material stored in chromosomes and this is the physical basis for the transmission of the characteristics of living organisms from one generation to the other. The term 'Gene' was introduced by Johannsen in 1909. Genes can be imagined as being strung together like beads to form the threadlike chromosomes. Genes can also predict diseases. A genetic linkage map was published in 1987 of the entire human genome covering the 23 pairs associated with genetically base human illnesses.

★ General Properties and Function:

More important properties and functions of the genes are described below:

(i) Capability to determine traits, e.g., Skin colour, intelligence, temperament,

blood group etc. They maintain the specificity of an individual.

(ii) Ability to undergo identical reproduction. They play vital role in the transmission of characteristics from parents to off-spring.

(iii) Ability to undergo mutation. This term is used to denote a physio-chemical change in genes which alter the effect on the character influenced by it.

(iv) Genes are very important for synthesis of various proteins and enzymes of the cell.

★ Classification of Genes

It can be classified into the following categories.

PUACP

Structural Gene

It controls the structure of 3 polypeptide comprising the protein.

Super Gene

It is a group of genes which encode a number of functionally related protein.

Operator Gene

It exercises control over a set of contiguous structural genes and collectively constitutes an operon.

Regulator Gene

It is responsible for controlling the particular activity of a specific operation.

Sex linked Gene

These genes are linked with Sex Chromosomes.

★ Chemical Basis:

DNA is the chemical basis of genes and is indeed the hereditary material. Genes act through controlling the protein constitution of cells.

★ Number:

Every gene lies in a particular position on the chromosomes to which it belongs. This specific position is HMI?M<:ii?BE1

referred to as the locus of Genes.

Nucleus of chromosome contains about 20,000 genes, half of which being contributed by each parent.

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Cytoplasm is a darker structure which has a rough surface. This is called the nucleus. This nucleus controls the anabolic activities and the cytoplasm controls the catabolic activities.

* Different Parts of Cell:

Essential Parts of a Cell are described below:

Cell membrane

Cell membrane

Small temporary vacuoles

Nucleus Cytoplasm

PUACP

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NUCLEIC ACIDS

Nucleic acids are complex biomolecules that contain the genetic instructions used in the development and function of all living organisms.

There are two main types of nucleic acid:

1. Deoxyribonucleic Acid (DNA)

DNA is a double-standed helix made up of nucleotides, each composed of a sugar molecule called deoxyribose, a phosphate group, and one of four nitrogenous bases:

- Adenine (A)
- Guanine (G)
- Cytosine (C)
- Thymine (T)



DNA is responsible for storing and transmitting genetic information from one generation to the next.

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2. Ribonucleic Acid (RNA)

RNA is a single-stranded molecule made up of nucleotides, each composed of a sugar molecule called ribose, a phosphate group, and one of four nitrogenous bases:

- Adenine (A)
- Guanine (G)
- Cytosine (C)
- Uracil (U)

RNA plays a crucial role in protein synthesis, transmitting genetic information from DNA to the ribosomes, where proteins are synthesized.

Protein

What is Protein?

Proteins are large complex molecules that are essential for the structure, function and regulation of the body's cells, tissues, and organs. They are made up of long chains of amino acids, which are organic compounds that combine in specific sequences to form proteins.

Structure of Protein

- ① **Primary structure:** ~~sequence of amino acids~~
Sequence of amino acids.
- ② **Secondary structure:** The folding or coiling of the amino acid chain into structures like alpha helices or beta sheets.
PUACP
- ③ **Tertiary structure:** The overall 3D shape of the protein formed by interaction between amino acid side chains.
- ④ **Quaternary structure:** The combination of multiple protein subunits to form a larger functional unit.

Function of Protein.

Protein have a wide variety of function in the body.

→ Building and repairing tissues:

Protein ^{are} key to the growth and repair of muscles, skin, and other tissues.

→ **Enzyme Production:** Proteins act as enzymes that speed up chemical reactions in the body.

→ **Immune support:** Antibodies, which are proteins, help protect the body from infections.

→ **Transportation:** Proteins like hemoglobin transport oxygen throughout the body.

Sources of Protein

Protein are found in a variety of food sources:

Animal-based sources: Meat,

Poultry, fish, egg, and dairy products.

Plant-based sources: Legumes, nuts.

Seeds, and grains like quinoa.

The
Collector

DNA

Deoxyribonucleic Acid (DNA) is the molecules that carries the genetic instructions for life. It is the fundamental substance that determines the inherited traits and characteristics of all living organisms, from bacteria to humans. DNA is often referred to as the "blueprint of life" because it contains the instructions for building and maintaining an organism.

Structure of DNA

① Double Helix:

DNA has a unique structure called the double helix which looks like a twisted ladder. The two strands of the helix are made of ^{log} chain of nucleotides.

2. Nucleotide:

DNA is composed of smaller units called

nucleotides. Each nucleotides consists of three parts: (DNA)

- ★ A sugar molecule & A phosphate group

- ★ A nitrogenous base (one of four:

adenine (A) thymine (T) cytosine (C) or guanine (G))

3 Base Pairing

The two strands of DNA are connected by base pairs. The bases follow specific pairing rules:

Adenine (A) Pairs with Thymine (T)

Cytosine (C) pairs with Guanine (G)

4. Antiparallel Strands:

The two strands of DNA run in opposite direction. This is called anti-parallel orientation. One strand runs in the 5' to 3' direction and the other runs in the 3' to 5' direction.

Function of DNA

1. Genetic Information Storage:

DNA stores the genetic instructions for the development growth and functioning of living organisms.

2. Protein Synthesis:

DNA provides the code for making proteins, which are essential for cell structure and function.

3. Replication:

DNA can replicate itself to ensure that each new cell receives an identical copy of genetic information during cell division.

2. Inheritance:

DNA is passed from parents to offspring, carrying genetic traits across generations.

HORMONES

Hormones are chemical messengers produced by various glands in the body, such as the endocrine glands (e.g., the thyroid, adrenal glands, and pancreas).

They are released into the bloodstream and travel to different tissues or organs, where they help regulate processes like metabolism, growth, mood, and reproduction. Hormones control a wide range of functions, including:

1. Metabolism

Hormones like insulin regulate blood sugar levels, while thyroid hormones control the rate of metabolism.

2. Growth and development

Growth hormones play a key role in physical growth, and sex hormones like estrogen and testosterone influence puberty and reproductive health.

2. Reproduction

Hormones regulate menstrual cycles, pregnancy and sperm production.

4. Mood and emotions

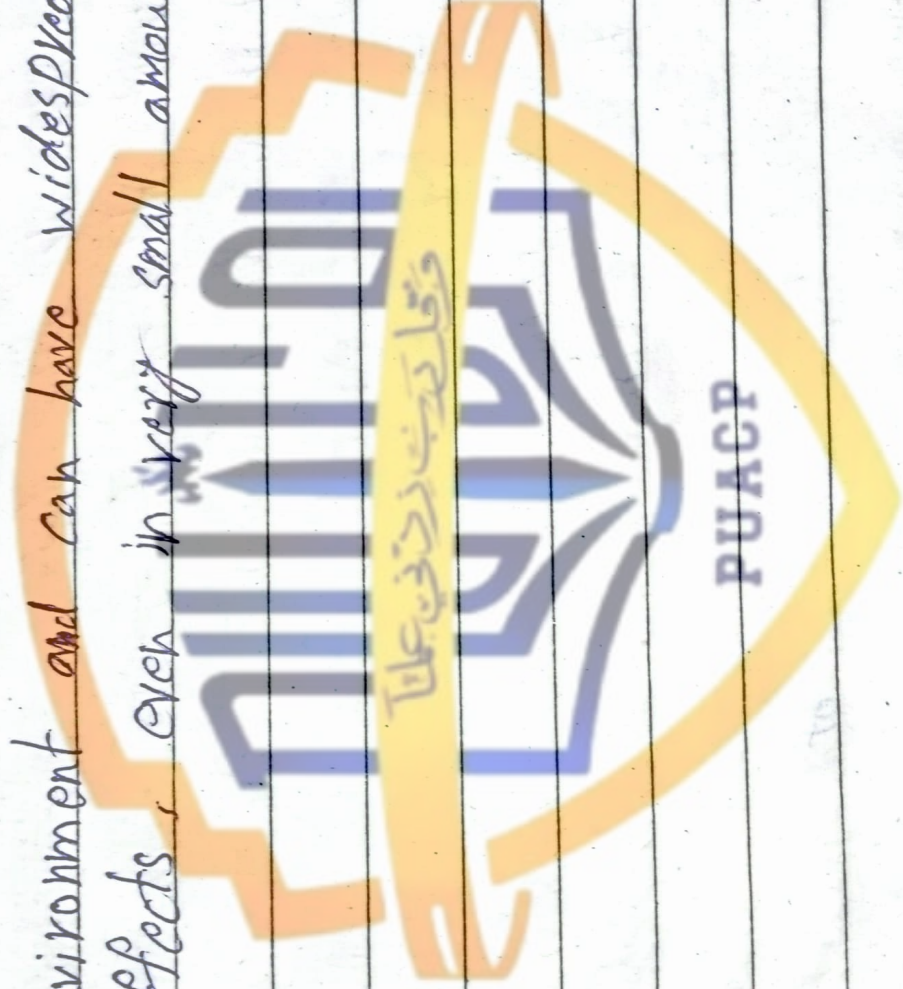
Hormones such as cortisol and

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Serotonin affect stress, mood, and overall emotional well-being.

They are produced in response to signals from the body and external environment and can have widespread effects, even in very small amounts.



Nutrient Concept of balance diet

The Concept of balance diet revolves around Consuming the right proportions of different types of nutrients to maintain optimal health and well being. A balanced diet provides the body with the essential nutrients it needs for energy, growth, repair, and proper functioning. ~~Proper~~

These nutrients include:

1 Carbohydrates

The body's Primary Source of energy, found in foods like bread, rice, pasta, fruits, and vegetables.

2 Proteins

Essential for growth, repair of tissues, and immune function. Good Sources include meat, fish, eggs, legumes, and nuts.

3. Fats

Provide a Concentrated Source of energy and are important for the absorption of fat-soluble vitamins (A, D, E, K). Healthy fats are found in olive oil, avocados, and fatty fish.

4. Vitamins

Vital for various biochemical functions. For example, Vitamin C supports immune function, and Vitamin D aids in calcium absorption. These are found in fruits, vegetables, and fortified foods.

5. Minerals

Crucial for various bodily functions. Such as bone health (Calcium), nerve function (Sodium), and oxygen transport (Iron). Key sources include dairy products, leafy greens, and meat.

6. Fiber

Important for digestive health and maintaining stable blood sugar levels. It's found in fruits, vegetables, whole grains, and legumes.

7. Water

Essential for hydration, digestion, and the regulation of body temperature.



Metabolism

Metabolism refers to all the chemical reactions that occur within the body to maintain life.

These reactions are essential for converting the food we eat into energy and for building and repairing tissues. Metabolism is generally divided into two main processes:

Anabolism

The process of building up or synthesizing molecules, such as proteins, fats and nucleic acids, from simple ones. Anabolism requires energy and is essential for growth, repair, and maintenance of the body's cells and tissues.

Catabolism

The breakdown of complex molecules into simpler ones, releasing energy. For example, the breakdown of carbohydrates into glucose

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or fats into fatty acids and glycerol provides the body with energy to perform various functions like movement, cell division, and maintaining body temperature.

Metabolism is influenced by factors such as age, gender, genetics, physical activity and diet.

The rate at which metabolism occurs is referred to as the metabolism rate. Which can vary from person to person. Metabolism help regulate

important fraction like 20

Liberty Production

Converting food into usable energy

Temperature regulation

Maintaining a stable internal body temperature.

Hormonal balance &

Regulation processes like hunger, growth, and stress responses.

